

Sungyeon Kook

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RESEARCH INTERESTS

Quantum error correction; fault-tolerant quantum computing; logical operations; decoders; code architectures

EDUCATION

Korea University (KU)

Seoul, Korea

M.S. in Electrical Engineering

Thesis: *Method for Implementing Low Spatial Cost CCZ Magic State Factory*

Mar. 2024 – Feb. 2026

B.S. in Electrical Engineering

Thesis: *Logical qubits based on surface code at quantum gate level*

Mar. 2020 – Feb. 2024

PUBLICATIONS AND MANUSCRIPTS

* First author(s); † Corresponding author.

- [1] **Sungyeon Kook***, Yujin Kang, Ilkwon Sohn, and Jun Heo†.
“Low Spatial Cost CCZ Magic State Factory.”
Under review at *Quantum Information Processing*. [[arXiv:2606.24170](#)].
- [2] Changyeol Lee*, **Sungyeon Kook***, Wooyeong Song, Kwangil Bae, Wonhyuk Lee, and Ilkwon Sohn†.
“Efficient Quantum Modular Reduction: Crandall Reduction and Its Fault-Tolerant Resource Analysis.”
Submitted to *Scientific Reports*. [[arXiv:2608.11563](#)].
- [3] Wooyeong Song*, **Sungyeon Kook***, Wonhyuk Lee, and Ilkwon Sohn†.
“Asymmetry-Aided Measurement-Based Quantum Repeaters and Distributed Quantum Computing with a Decoder-Free Client.”
Submitted to *npj Quantum Information*. [[arXiv:2607.11238](#)].
- [4] **Sungyeon Kook***, Yujin Kang, and Jun Heo.
“Limited Connectivity Quantum Architectures: Enhancing Surface Code Implementation Through Routing Qubits.”
15th International Conference on Information and Communication Technology Convergence (ICTC),
pp. 762–763, 2024.
- [5] Yujin Kang*, Youshin Chung, Huidan Zheng, **Sungyeon Kook***, and Jun Heo.
“Magic State Distillation with Reduced Time Cost.”
IEEE International Conference on Quantum Computing and Engineering (QCE), vol. 2, pp. 416–417, 2024.

RESEARCH CONTRIBUTIONS

Low-Spatial-Cost CCZ Magic-State Factory [1]

2024 – Present

Advisor: Prof. [Jun Heo](#)

Korea University

- Converted the $[[8, 3, 2]]$ circuit for CCZ magic-state distillation into eight Z -basis $\pi/8$ Pauli-product joint measurements, eliminating eight $|0\rangle$ auxiliary qubits and reducing the factory’s persistent logical-qubit requirement from 12 to 4.
- Established single-fault detection for the joint-measurement construction and derived error models for two-level magic-state distillation, accounting for both surface-code logical faults during factory operation and errors in the injected magic states.
- Designed a compact surface-code implementation using four-qubit twisted ZY joint measurements; quantified 71.4% lower spatial cost and 37.7% lower space-time volume for two-level CCZ magic-state distillation under pipelined scheduling.

Fault-Tolerant Code Switching and Decoder-Free Quantum Networking [3]

2024 – Present

Advisors: Prof. [Jun Heo](#) and Dr. [Wooyeong Song](#)

Korea University and KISTI

- Developed multi-round gauge fixing between asymmetric Bacon–Shor and rotated surface codes, preserving logical Pauli algebra and achieving distance- m switching for $BS(m \times 3) \rightarrow SC(m \times m)$ versus distance 1 for switching.
- Constructed a centralized decoding framework to jointly process syndrome information from Bacon–Shor transmission, surface-code computation, and return transmission, enabling global QEC without real-time client decoding.

Fault-Tolerant Resource Analysis for Quantum Modular Reduction [2]

2026 – Present

Advisor: Dr. [Ilkwon Sohn](#)

KISTI

- Derived circuit-specific logical-error budgets from qubit counts and T -depth using the KQ formalism; determined required rotated surface-code distances for modular-reduction circuits at specified circuit failure probabilities and physical error rates.
- Formulated a runtime model combining T -state distillation and decoder backlog for modular reduction, quantifying how T -depth reductions affect code distance, magic-state latency, classical decoding workload, and total execution time.

Architecture-Aware Quantum Circuit Mapping and Scheduling

2024 – 2025

Advisor: Prof. [Jun Heo](#)

Korea University

- Analyzed trade-offs between flag-qubit resources and SWAP operations for surface-code mapping on heavy-hexagon architectures; formulated a QODG-based scheduler for circuit execution on checkerboard lattice-surgery architectures.
- Developed a Greedy DAG scheduling pass in Qiskit that constructs parallel gate layers by selecting operations whose dependencies are satisfied and whose qubit requirements do not conflict, exposing additional gate-level parallelism.

ADDITIONAL RESEARCH EXPERIENCE

Crosstalk-Aware Surface-Code Scheduling via Graph Coloring

2024 – 2025

Advisor: Prof. [Jun Heo](#) | [National R&D Project](#)

Korea University

- Formulated CNOT incompatibilities from superconducting crosstalk as a graph-coloring problem to identify groups of compatible stabilizer-measurement operations for parallel execution; scheduled the stabilizer CNOT network of a distance-24 surface-code circuit at depth 8, using IBM-inspired hardware parameters while satisfying the imposed crosstalk constraints.

Experimental Quantum Error Correction on Superconducting Qubits

2024 – 2025

Advisor: Dr. [Hwan-Seop Yeo](#) | [National R&D Project](#)

KRISS

- Characterized superconducting qubits using spectroscopy, T_1/T_2 measurements, DRAG calibration, and randomized benchmarking to extract hardware parameters for QEC experiments; developed connectivity-aware CNOT/SWAP operations and experimentally implemented stabilizer-projection circuits, validating them through quantum state tomography.

Variational Quantum Algorithms

2023

Advisor: Prof. [Mahn-Soo Choi](#)

Korea University

- Implemented VQE and QAOA circuits in Qiskit and PennyLane, using classical optimization to tune circuit parameters; proposed a mutual-information-based quantum kernel function to capture correlations between input features.

SELECTED CONFERENCE CONTRIBUTIONS

Oral presentations

- Joint Conference on Communications and Information (JCCI 2025) Apr. 2025
- International Conference on ICT Convergence (ICTC 2024)** [\[link\]](#) Oct. 2024

Poster presentations

- 2026 Annual Meeting of the Quantum Information Society of Korea Feb. 2026
- International Conference on ICT Convergence (ICTC 2025)** [\[link\]](#) Oct. 2025
- Quantum Error Correction Conference (QEC 2025)** [\[link\]](#) Aug. 2025

Co-author contributions

- Asian Quantum Information Science Conference (AQIS 2026)** [\[link\]](#) Aug. 2026
- Optics and Photonics Congress (OPC 2026) Jul. 2026
- IEEE International Conference on Quantum Computing and Engineering (QCE 2024)** [\[link\]](#) Sep. 2024

RESEARCH APPOINTMENTS

- Korea Institute of Science and Technology Information (KISTI)** — Post-Master Researcher Mar. 2026 – Present
Quantum Network Research Center
- Korea University (KU)** — Graduate Research Assistant Mar. 2024 – Feb. 2026
Communications and Information Systems Laboratory
- Korea Research Institute of Standards and Science (KRISS)** — Visiting Researcher Aug. 2024 – Aug. 2025
Center for Superconducting Quantum Computing System
- Korea University (KU)** — Undergraduate Research Assistant Jan. 2023 – Feb. 2024
Communications and Information Systems Laboratory
Jul. 2023 – Aug. 2023
Quantum Control Laboratory

TECHNICAL SKILLS

Programming: Python, C, MATLAB **Quantum software:** Qiskit, Stim, PyMatching, PennyLane

TEACHING AND PROFESSIONAL ACTIVITIES

- IBM Qiskit Advocate** [\[link\]](#) Aug. 2025 – Present
- Teaching Assistant, Data Science and Artificial Intelligence Fall 2024
- Co-Founder and Treasurer, QUantum Information Club at Korea University (QUICK)** Jan. – Dec. 2023
- Organizer, Qiskit Fall Festival 2023** [\[link\]](#) Oct. 2023

HONORS AND AWARDS

- Outstanding Academic Achievement**, Korea University Spring 2023
- Excellence Award (Korea University President's Award)**, Creative Challenger Program 2022
- Encouragement Award, Startup Express Summer Session, Korea University 2022

REFERENCES

- Prof. Jun Heo** [\[Faculty\]](#) [\[Google Scholar\]](#) [\[Website\]](#) [\[IEEE\]](#) *School of Electrical Engineering*
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- Prof. Joongheon Kim** [\[Faculty\]](#) [\[Google Scholar\]](#) [\[ORCID\]](#) [\[Website\]](#) [\[IEEE\]](#) *School of Electrical Engineering*
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